FACULTE DES SCIENCES ET TECHNOLOGIES

Sciences Informatiques

Cours : Progrmmation Python

TD Nº 3 : Python

Blomy ANTOINE

L3

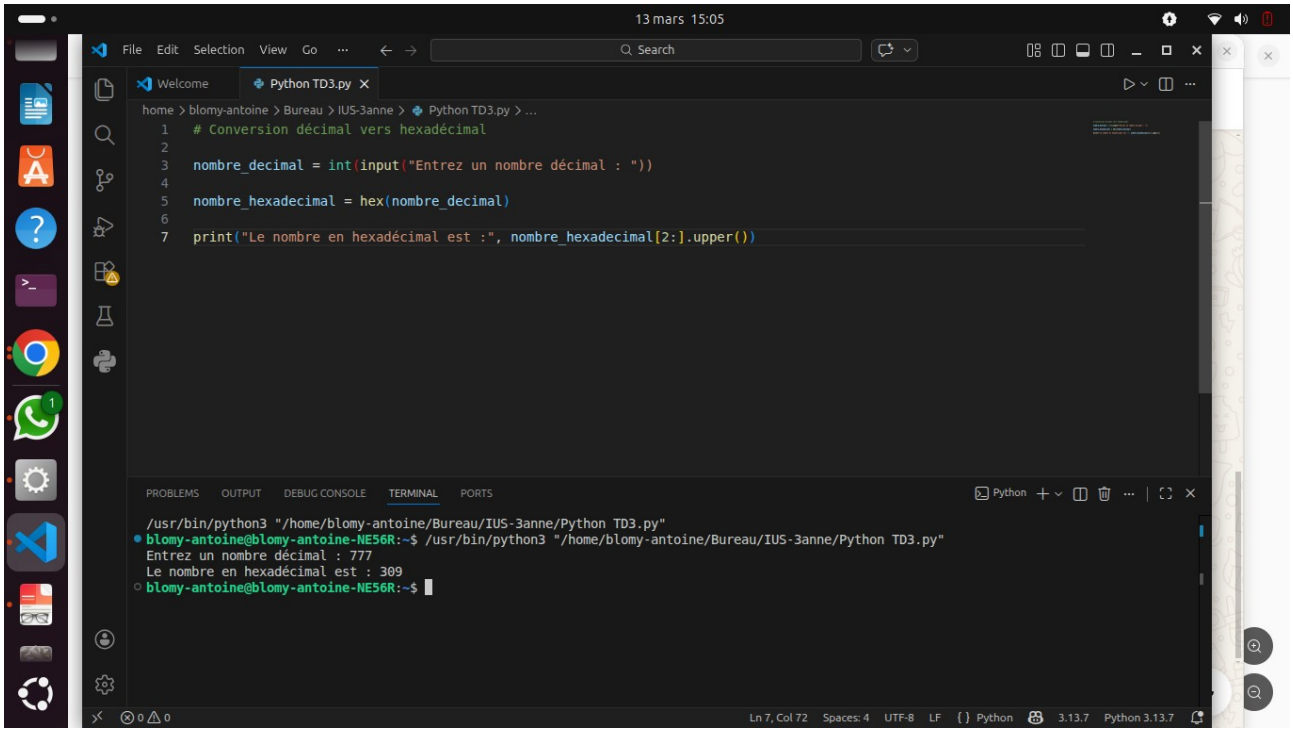
Sous la direction du professeur :
Ismaël SAINT-AMOUR

13 MARS 2026

1- J’écris un programme qui convertit un nombre décimal en hexadécimal en demandant à l'utilisateur de saisir ce nombre.

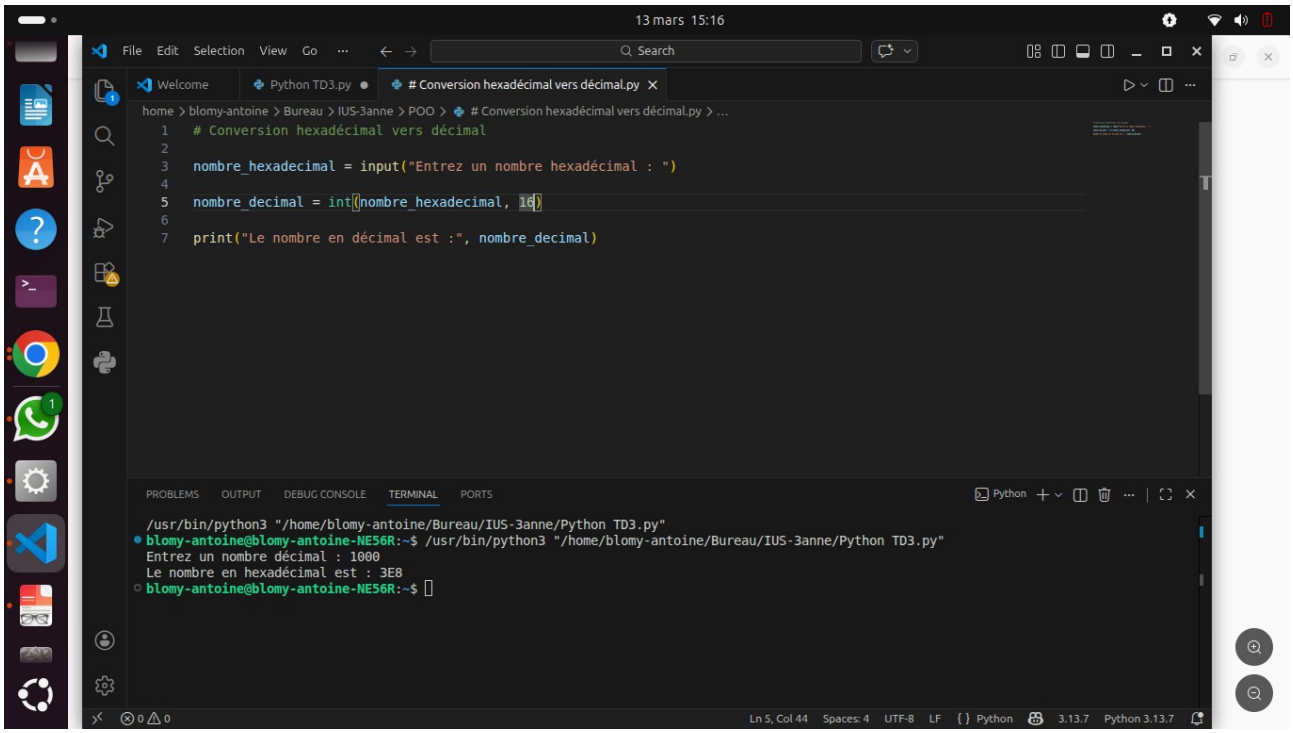
Programme Python : Conversion d’un nombre décimal en hexadécimal

Le programme suivant demande à l’utilisateur de saisir un nombre décimal puis affiche sa conversion en hexadécimal.



2- Programme Python : Conversion d'un nombre hexadécimal en décimal

Ce programme permet de convertir un nombre hexadécimal saisi par l'utilisateur en

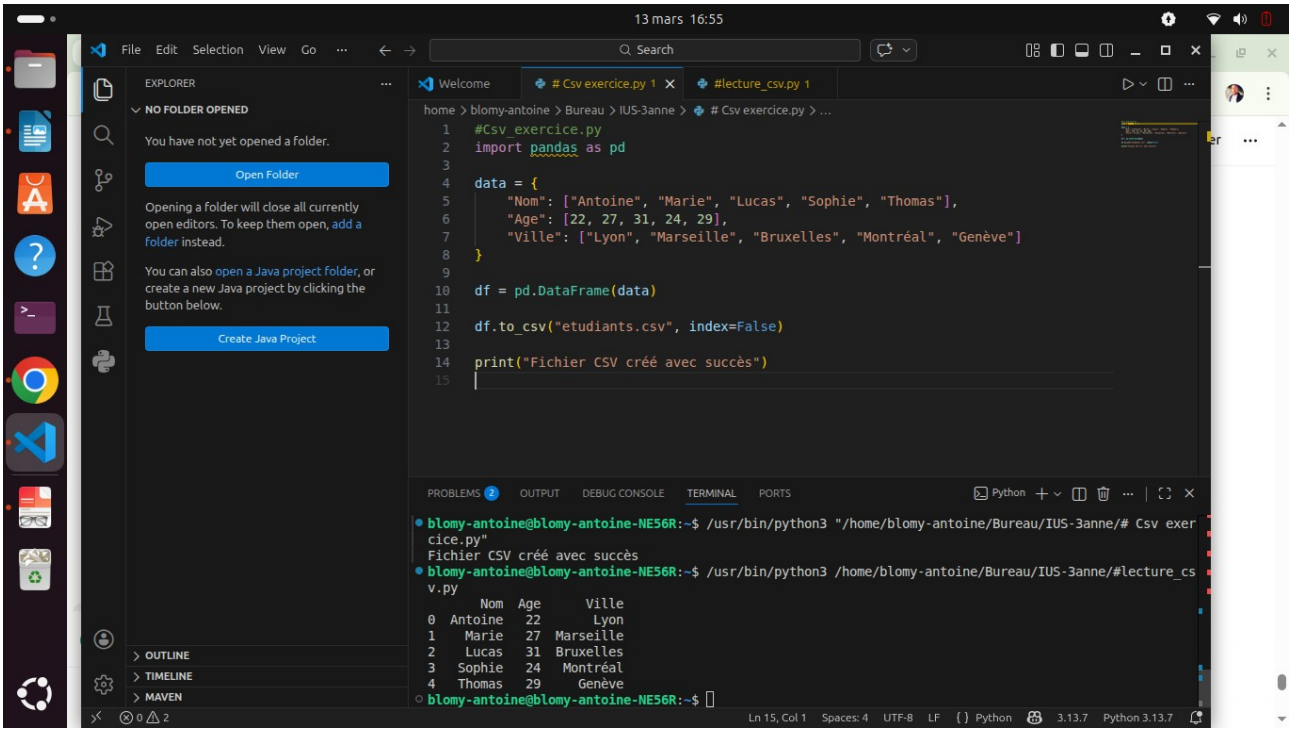


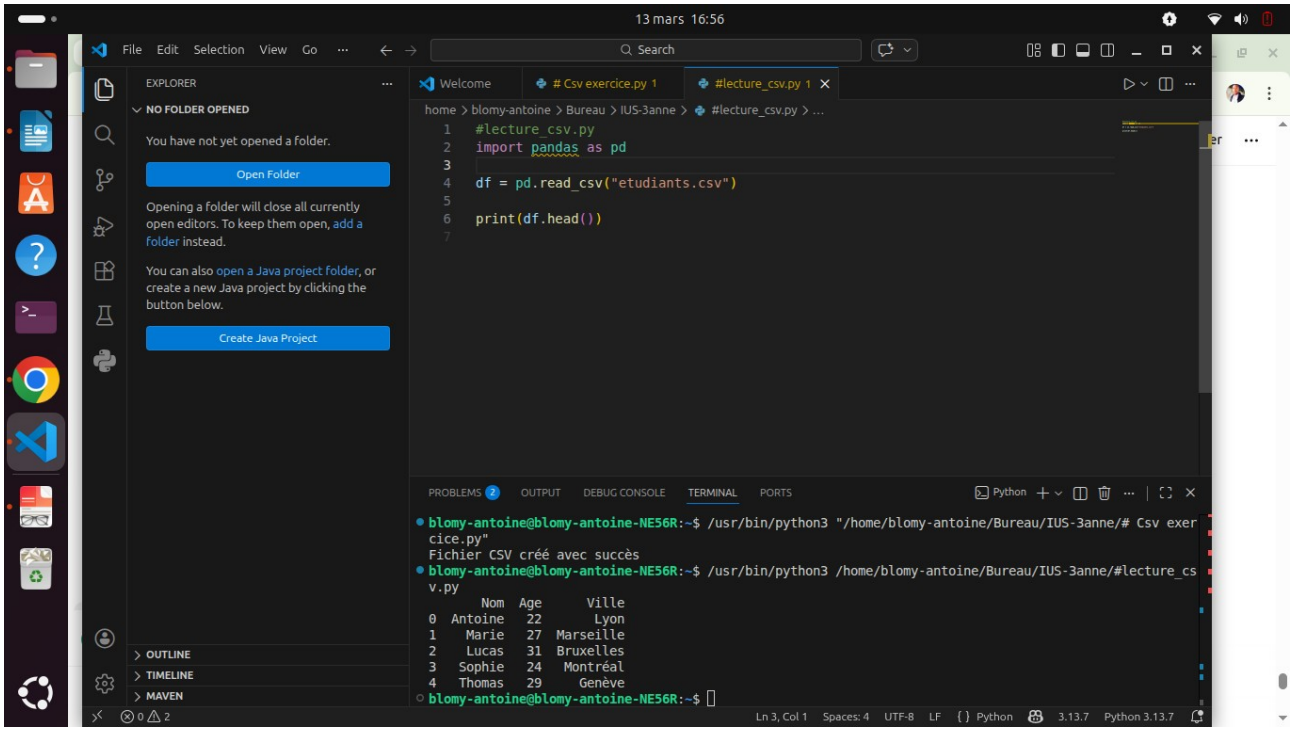
nombre décimal.

3- Création et lecture d'un fichier CSV avec Python

Les fichiers CSV sont très utilisés pour stocker et manipuler des données.

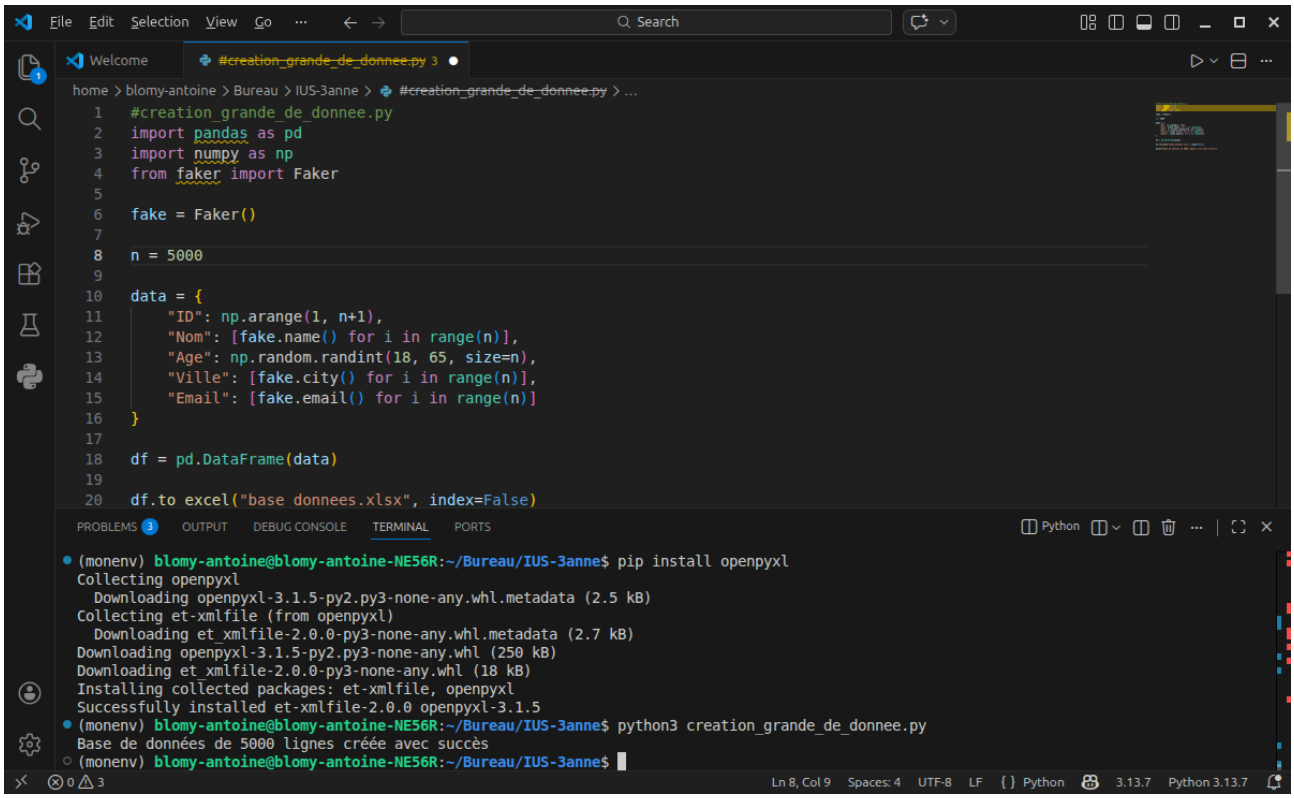
Création du fichier CSV


13 mars 16:55
File Edit Selection View Go
Search
EXPLORER
NO FOLDER OPENED
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
You can also open a Java project folder, or create a new Java project by clicking the button below.
Create Java Project
OUTLINE
TIMELINE
MAVEN
Welcome
Csv exercice.py 1
#lecture_csv.py 1
home > blomy-antoine > Bureau > IUS-3anne > # Csv exercice.py > ...
#Csv_exercice.py
import pandas as pd
data = {
"Nom": ["Antoine", "Marie", "Lucas", "Sophie", "Thomas"],
"Age": [22, 27, 31, 24, 29],
"Ville": ["Lyon", "Marseille", "Bruxelles", "Montréal", "Genève"]
}
df = pd.DataFrame(data)
df.to_csv("etudiants.csv", index=False)
print("Fichier CSV créé avec succès")
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
blomy-antoine@blomy-antoine-NE56R:~$ /usr/bin/python3 "/home/blomy-antoine/Bureau/IUS-3anne/# Csv exercice.py"
Fichier CSV créé avec succès
blomy-antoine@blomy-antoine-NE56R:~$ /usr/bin/python3 /home/blomy-antoine/Bureau/IUS-3anne/#lecture_csv.py
Nom Age Ville
0 Antoine 22 Lyon
1 Marie 27 Marseille
2 Lucas 31 Bruxelles
3 Sophie 24 Montréal
4 Thomas 29 Genève
blomy-antoine@blomy-antoine-NE56R:~$
Ln 15, Col 1 Spaces: 4 UTF-8 LF { } Python 3.13.7 Python 3.13.7



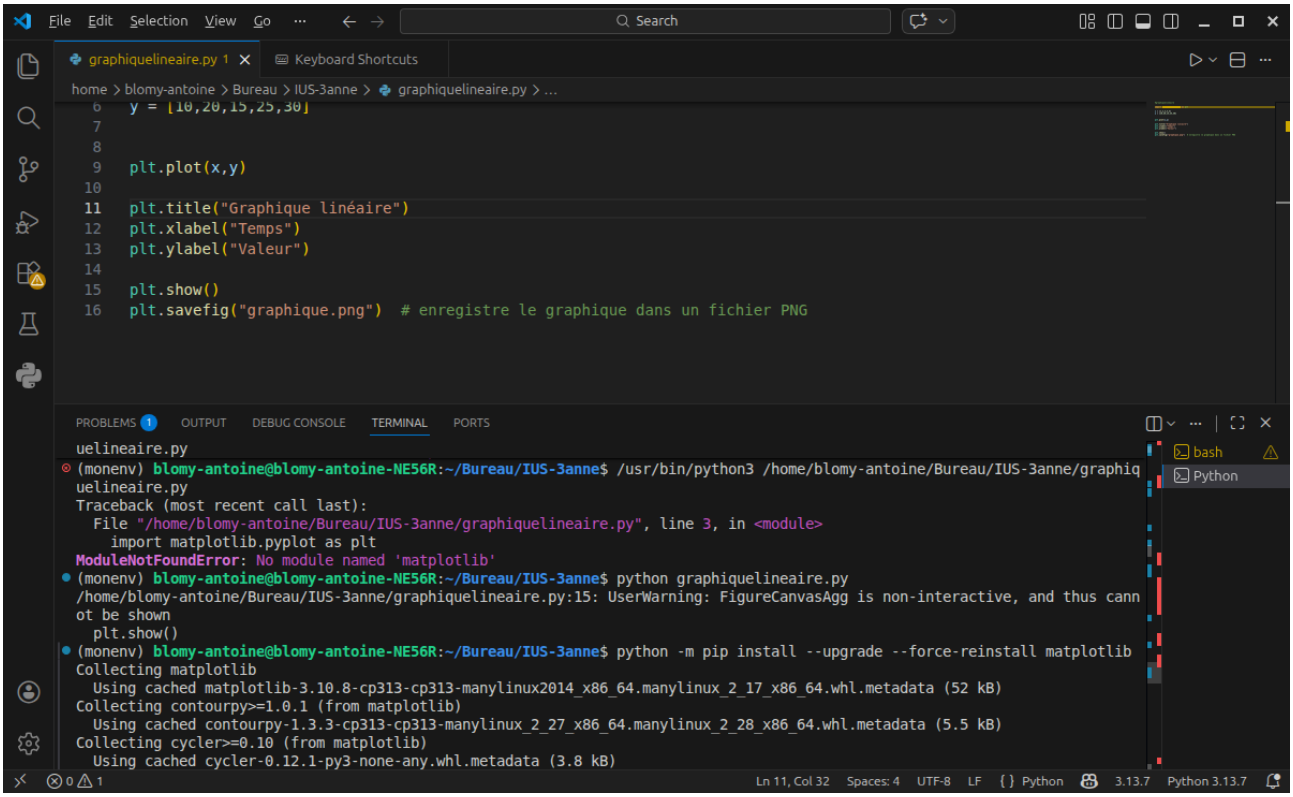
4- Création d'une grande base de données (5000 lignes)

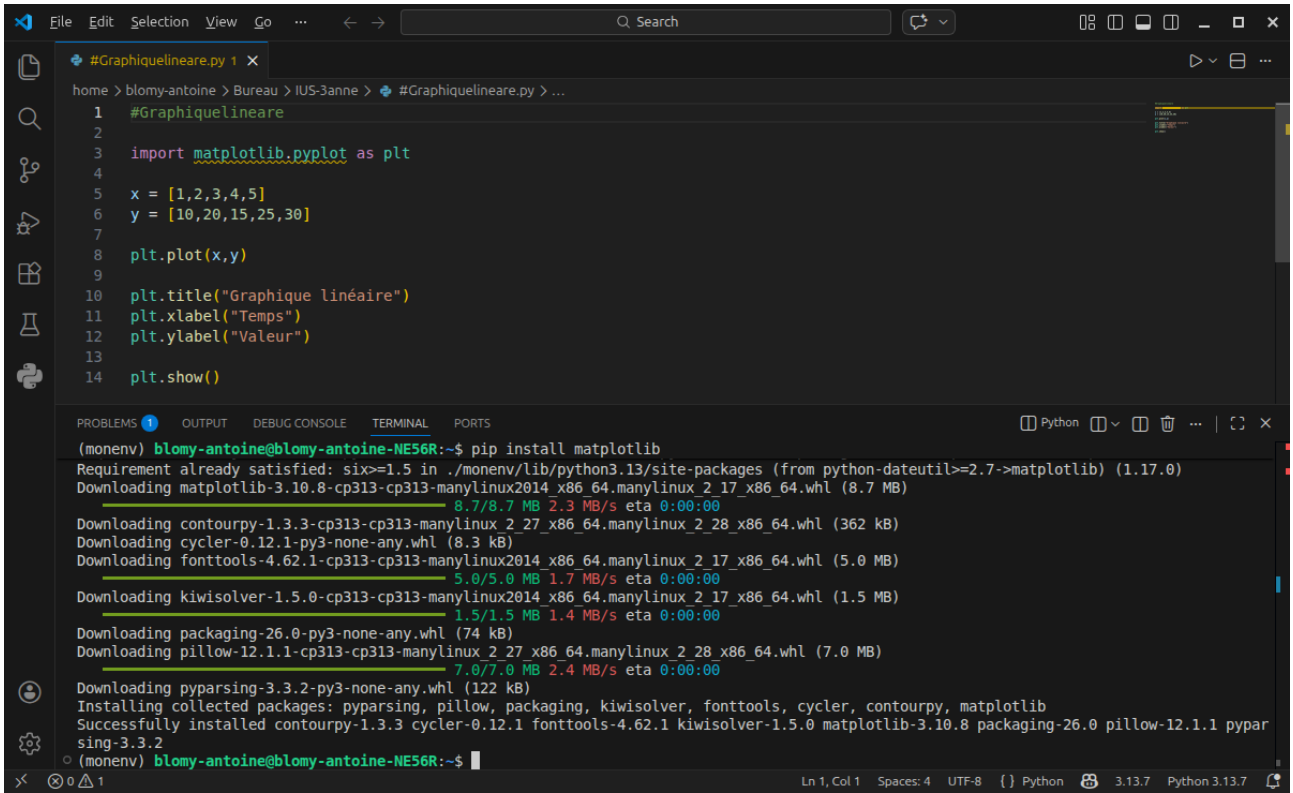
Ce programme génère automatiquement une base de données contenant 5000 lignes.



5- Création de trois graphes avec Matplotlib

Graphique 1 : Graphique linéaire


File Edit Selection View Go
Search
graphiquelineaire.py 1
Keyboard Shortcuts
home > blomy-antoine > Bureau > IUS-3anne > graphiquelineaire.py > ...
y = [10,20,15,25,30]
plt.plot(x,y)
plt.title("Graphique linéaire")
plt.xlabel("Temps")
plt.ylabel("Valeur")
plt.show()
plt.savefig("graphique.png") # enregistre le graphique dans un fichier PNG
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
uelineaire.py
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ /usr/bin/python3 /home/blomy-antoine/Bureau/IUS-3anne/graphiq
uelineaire.py
Traceback (most recent call last):
File "/home/blomy-antoine/Bureau/IUS-3anne/graphiquelineaire.py", line 3, in <module>
import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ python graphiquelineaire.py
/home/blomy-antoine/Bureau/IUS-3anne/graphiquelineaire.py:15: UserWarning: FigureCanvasAgg is non-interactive, and thus cann
ot be shown
plt.show()
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ python -m pip install --upgrade --force-reinstall matplotlib
Collecting matplotlib
Using cached matplotlib-3.10.8-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (52 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
Using cached contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
bash
Python
Ln 11, Col 32 Spaces: 4 UTF-8 LF {} Python 3.13.7 Python 3.13.7


File Edit Selection View Go
Search
#Graphiquelineare.py 1
home > blomy-antoine > Bureau > IUS-3anne > #Graphiquelineare.py > ...
#Graphiquelineare
import matplotlib.pyplot as plt
x = [1,2,3,4,5]
y = [10,20,15,25,30]
plt.plot(x,y)
plt.title("Graphique linéaire")
plt.xlabel("Temps")
plt.ylabel("Valeur")
plt.show()
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
(monenv) blomy-antoine@blomy-antoine-NE56R:~$ pip install matplotlib
Requirement already satisfied: six>=1.5 in ./monenv/lib/python3.13/site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.10.8-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7 MB)
8.7/8.7 MB 2.3 MB/s eta 0:00:00
Downloading contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (362 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.62.1-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.0 MB)
5.0/5.0 MB 1.7 MB/s eta 0:00:00
Downloading kiwisolver-1.5.0-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.5 MB)
1.5/1.5 MB 1.4 MB/s eta 0:00:00
Downloading packaging-26.0-py3-none-any.whl (74 kB)
Downloading pillow-12.1.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (7.0 MB)
7.0/7.0 MB 2.4 MB/s eta 0:00:00
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, packaging, kiwisolver, fonttools, cycler, contourpy, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 kiwisolver-1.5.0 matplotlib-3.10.8 packaging-26.0 pillow-12.1.1 pypar
sing-3.3.2
(monenv) blomy-antoine@blomy-antoine-NE56R:~$
Ln 1, Col 1 Spaces: 4 UTF-8 {} Python 3.13.7 Python 3.13.7

graphiquelineaire.py 1 | Keyboard Shortcuts

home > blomy-antoine > Bureau > IUS-3anne > graphiquelineaire.py > ...

```python
y = [10,20,15,25,30]


plt.plot(x,y)

plt.title("Graphique linéaire")
plt.xlabel("Temps")
plt.ylabel("Valeur")

plt.show()
plt.savefig("graphique.png")  # enregistre le graphique dans un fichier PNG
```

PROBLEMS 1 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
uelineaire.py
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ /usr/bin/python3 /home/blomy-antoine/Bureau/IUS-3anne/graphiq
uelineaire.py
Traceback (most recent call last):
  File "/home/blomy-antoine/Bureau/IUS-3anne/graphiquelineaire.py", line 3, in <module>
    import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ python graphiquelineaire.py
/home/blomy-antoine/Bureau/IUS-3anne/graphiquelineaire.py:15: UserWarning: FigureCanvasAgg is non-interactive, and thus cann
ot be shown
  plt.show()
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ python -m pip install --upgrade --force-reinstall matplotlib
Collecting matplotlib
  Using cached matplotlib-3.10.8-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (52 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Using cached contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
```

Ln 11, Col 32 | Spaces: 4 | UTF-8 | LF | { } Python | 3.13.7 | Python 3.13.7

graphiquelineaire.py 1

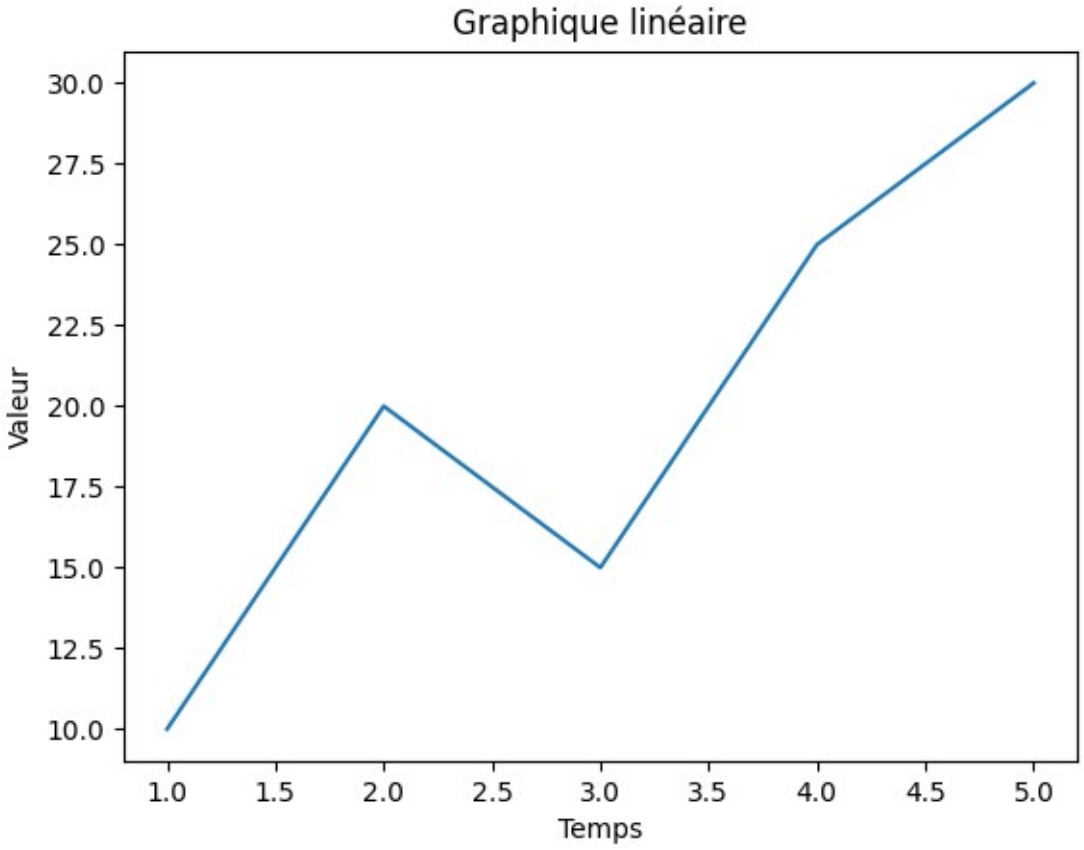
home > blomy-antoine > Bureau > IUS-3anne > graphiquelineaire.py > ...

```python
x = [1,2,3,4,5]
y = [10,20,15,25,30]


plt.plot(x,y)

plt.title("Graphique linéaire")
plt.xlabel("Temps")
plt.ylabel("Valeur")

plt.show()
```

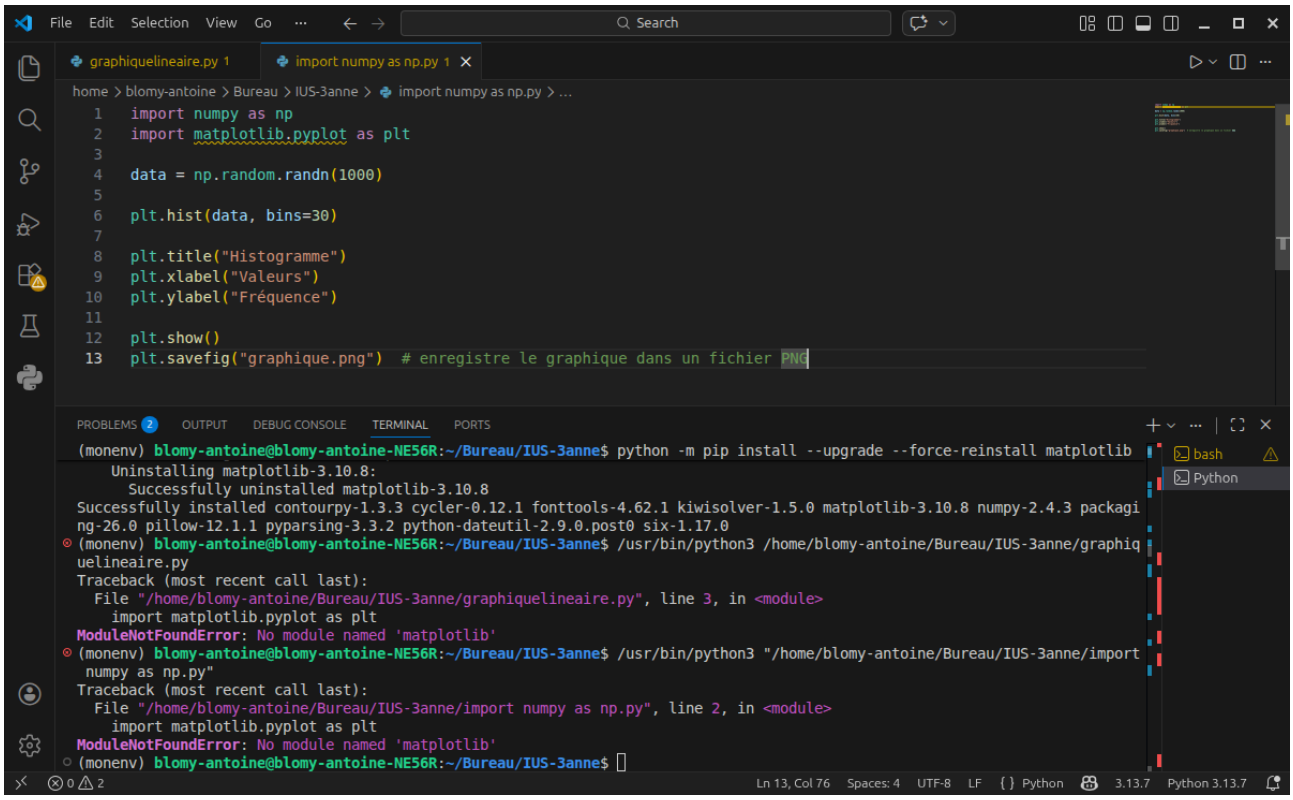
PROBLEMS 1 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
(monenv) blomy-antoine@blomy-antoine-NE56R:~$ pip install pandas openpyxl Faker matplotlib
Collecting et-xmlfile (from openpyxl)
  Using cached et_xmlfile-2.0.0-py3-none-any.whl.metadata (2.7 kB)
Requirement already satisfied: contourpy>=1.0.1 in ./monenv/lib/python3.13/site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in ./monenv/lib/python3.13/site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in ./monenv/lib/python3.13/site-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in ./monenv/lib/python3.13/site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in ./monenv/lib/python3.13/site-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in ./monenv/lib/python3.13/site-packages (from matplotlib) (12.1.1)
Requirement already satisfied: pyparsing>=3 in ./monenv/lib/python3.13/site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in ./monenv/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas) (1.17
.0)
Using cached openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)
Using cached faker-40.11.0-py3-none-any.whl (2.0 MB)
Using cached et_xmlfile-2.0.0-py3-none-any.whl (18 kB)
Installing collected packages: Faker, et-xmlfile, openpyxl
Successfully installed Faker-40.11.0 et-xmlfile-2.0.0 openpyxl-3.1.5
(monenv) blomy-antoine@blomy-antoine-NE56R:~$
```

Ln 6, Col 21 | Spaces: 4 | UTF-8 | LF | { } Python | 3.13.7 | Python 3.13.7



Graphique 2 : Histogramme


File Edit Selection View Go
Search
graphiquelineaire.py 1
import numpy as np.py 1
home > blomy-antoine > Bureau > IUS-3anne > import numpy as np.py > ...
import numpy as np
import matplotlib.pyplot as plt
data = np.random.randn(1000)
plt.hist(data, bins=30)
plt.title("Histogramme")
plt.xlabel("Valeurs")
plt.ylabel("Fréquence")
plt.show()
plt.savefig("graphique.png") # enregistre le graphique dans un fichier PNG
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ python -m pip install --upgrade --force-reinstall matplotlib
Uninstalling matplotlib-3.10.8:
Successfully uninstalled matplotlib-3.10.8
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 kiwisolver-1.5.0 matplotlib-3.10.8 numpy-2.4.3 packaging-26.0 pillow-12.1.1 pyparsing-3.3.2 python-dateutil-2.9.0.post0 six-1.17.0
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ /usr/bin/python3 /home/blomy-antoine/Bureau/IUS-3anne/graphiquelineaire.py
Traceback (most recent call last):
File "/home/blomy-antoine/Bureau/IUS-3anne/graphiquelineaire.py", line 3, in <module>
import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$ /usr/bin/python3 "/home/blomy-antoine/Bureau/IUS-3anne/import numpy as np.py"
Traceback (most recent call last):
File "/home/blomy-antoine/Bureau/IUS-3anne/import numpy as np.py", line 2, in <module>
import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'
(monenv) blomy-antoine@blomy-antoine-NE56R:~/Bureau/IUS-3anne$
bash
Python
Ln 13, Col 76 Spaces: 4 UTF-8 LF {} Python 3.13.7 Python 3.13.7

Conclusion

Ce travail dirigé a permis de se familiariser avec les bases de la **science des données avec Python**.
Les exercices réalisés ont permis de :

- installer les outils nécessaires (Python)
- comprendre les conversions entre différents systèmes de numération,
- manipuler des fichiers CSV,
- générer une base de données automatique,
- visualiser les données à l'aide de graphiques.

Ces compétences constituent une base importante pour l'analyse de données et l'apprentissage de techniques plus avancées en science des données.